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OVERVIEW

A systematic review on detection and adaptation of concept drift in streaming data using machine learning techniques

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Abstract

Last decade demonstrate the massive growth in organizational data which keeps on increasing multi-fold as millions of records get updated every second. Handling such vast and continuous data is challenging which further opens up many research areas. The continuously flowing data from various sources and in real-time is termed as streaming data. While deriving valuable statistics from data streams, the variation that occurs in data distribution is called concept drift. These drifts play a significant role in a variety of disciplines, including data mining, machine learning, ubiquitous knowledge discovery, quantitative decision theory, and so forth. As a result, a substantial amount of research is carried out for studying methodologies and approaches for dealing with drifts. However, the available material is scattered and lacks guidelines for selecting an effective technique for a particular application. The primary novel objective of this survey is to present an understanding of concept drift challenges and allied studies. Further, it assists researchers from diverse domains to accommodate detection and adaptation algorithms for concept drifts in their applications. Overall, this study aims to contribute to deeper insights into the classification of various types of drifts and methods for detection and adaptation along with their key features and limitations. Furthermore, this study also highlights performance metrics used to evaluate the concept drift detection methods for streaming data. This paper presents the future research scope by highlighting gaps in the existing literature for the development of techniques to handle concept drifts.

This article is categorized under:

Algorithmic Development > Ensemble Methods
Application Areas > Data Mining Software Tools
Fundamental Concepts of Data and Knowledge > Big Data Mining

KEYWORDS

concept drift, ensemble learning, incremental learning, machine learning, online learning, streaming data

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1 | INTRODUCTION

The technological evolution of internet of things (IoT) devices, smart devices, dynamic networks, market data, and so forthgenerate a massive volume of real-time data streams. The real-time data generated from multiple sources is known as heterogeneous streaming data. The process of transmitting a continuous flow of data (also known as data streams) to stream processing framework for extracting useful insights in near real-time is known as online stream processing. In the data stream applications, the instances arrive in a sequential fashion and at time-varying intervals. Therefore, it is uncertain how much data remains in the sequence because the data stream is not finite. The generated data of varying-intervals must be reviewed in close to real-time because one rarely gets a chance to get a brief of it once it is lost. Due to three main characteristics of data streams, that is, size, speed and data distribution, most data mining models are incompatible with data stream mining.

Stream processing is a computing paradigm that reads data in a stream from a collection of software or hardware sensors and evaluates continuous data streams, with feedback outputs. In a data stream, the data samples are produced linearly and autonomously following some probability distribution. Hence, it is crucial to detect any change in their distribution. There are some challenges associated with detection of change in data distribution over data streams. The techniques to identify changes in relationships within data is known as concept drift detection. These could be moderate changes over time, regular or periodic changes, such as that of seasonal data, or large and abrupt shifts. The COVID-19 lockdowns, for example, had a considerable impact on customer engagement and therefore, affecting the inventory management system for almost every manufacturing industry.

In the absence of annotated data, concept drift is often detected by examining the distribution of data instances. While dealing with real-world data stream problems, correct labels are frequently unavailable or almost inaccessible for concept drift detection. Further, the data distribution change could be classified as abrupt, gradual, or periodic. When the data distribution changes abruptly and there is no temporal gap between data instances, this is referred to as abrupt drift. Gradual drift is defined as an overlap between two concepts that have changed. Recurring drift refers to the phenomenon of seeing the same concept repeatedly over an extended period of time. Sudden drift can be easily detected as compared to gradual and recurrent drift.

1.1 | Summary of existing surveys in the field of concept drift detection

Hoens et al. (2012) is one of the earliest surveys related to streaming data in which the authors have explained the challenges of handling concept drifts and also outlined recent research for the development of a holistic framework. Furthermore, they also presented a comprehensive overview of difficulty with class disparity in streaming data. Gama et al. (2014) provided a survey based on drift adaptive learning techniques, an outline of evaluation processes, and practical applications for handling concept drift. It revolves around online labeled data streams where the relationship between the input target variable alters with time. Kadwe and Suryawanshi (2015) described a number of methods to deal with concept drifts. Applications of synthetic and real datasets with various concept drifts are also presented. Gomes et al. (2017) suggested a taxonomy for ensemble learning as a data streams based solution for an analysis of more than 60 techniques. An in-depth discussion of crucial elements, such as, diversity, is also carried out. Krawczyk et al. (2017) presented a literature review of ensembles for regression. Besides showcasing a broad range of ensemble techniques for streaming data, dynamic data learning ideas like Novelty detection, Inequalities in the data streams, and semi-supervised and active learning were also discussed. Iwashita and Papa (2019) reviewed several papers that address concept drift and provided a thorough analysis of available synthetic and real-world datasets that can be utilized to evaluate the concept drift detection algorithms. Additionally, they included a complete analysis of types and methods available for dealing with concept drifts.

Khamassi et al. (2018) presented a review of concept drift challenges and associated domains to assist multi-disciplinary researchers for taking concept drift into consideration for deployments. Wares et al. (2019) conducted a thorough analysis of concept drift in the context of stream mining. The authors also examined current challenges of evaluation process of concept drift detectors, such as the scarcity of existing datasets and the effect of temporal constraints on concept drift detection. They also provided suggestions based on the pros and cons of the algorithms. Gemaque et al. (2020) presented a detailed review of methods for dealing with unsupervised classification challenge of concept drift. Further, they also proposed a taxonomy of cutting-edge methods for concept drift detection based on unsupervised tactics. Agrahari and Singh (2021) conducted a survey on concept drift detectors with their main attractions and limitations. The main aim of their work is to outline research trends and work that can be done in this

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domain in the future. Fahy et al. (2023) presented an outline of the attractive ideas in this domain with recent advancements. They also included research trends, literature gaps, and future directions in the field of data streams. Suárez-Cetrulo et al. (2023) examined the latest advancements in data stream learning and machine learning techniques to address both non-stationary and repeating alterations in the generative process of a time series, with a particular focus on the subsequent assemblage of drifts. Sakurai et al. (2023) established a standard for four drift detection algorithms (EDDM, DDM, HDDMW, and HDDMA) that accurately identifies abrupt, gradual, and incremental drift types. In order to clarify the capabilities and potential trade-offs associated with choosing a concept drift method, the authors assessed some parameters like detection capacity, time, and delay.

In the recent decade, concept drift has been a commonly researched subject in the domain of streaming data, with comprehensive surveys on concept drift detection methods. However, the issue is that the available literature is fairly fragmented, and comparable concepts are produced under various taxonomies. In addition to an analysis of existing state-of-the-art approaches, this study aims to elicit discussion and assist readers in the identification of relevant algorithms suitable for their application scenarios. The current state-of-the-art detection methods are categorized uniquely in this article along with criticisms, future trends, and unresolved issues. The work done in this survey is highlighted in Section 1.2. Further, Table 1 compares previous surveys with our study on the basis of research questions mentioned in Table 2.

TABLE 1 Summary of survey papers on concept drift detection and adaptation in streaming data.

	Research questions									
Related surveys	**RQ1**	**RQ2**	**RQ3**	**RQ4**	**RQ5**	**RQ6**	**RQ7**	**RQ8**	**RQ9**	**RQ10**
Cetrulo et al. (Suárez-Cetrulo et al., 2023)	✓	✓	×	✓	✓	×	✓	✓	✓	✓
Fahy et al. (Fahy et al., 2023)	✓	✓	✓	×	×	×	×	✓	✓	✓
Agrahari et al. (Agrahari & Singh, 2021)	✓	✓	✓	×	×	×	×	✓	✓	✓
Hashmani et al. (Hashmani et al., 2020)	✓	✓	✓	×	×	×	✓	×	×	×
Gemaque et al. (Gemaque et al., 2020)	✓	✓	×	×	×	×	×	×	✓	×
Hu et al. (Hu et al., 2020)	✓	✓	×	×	×	×	×	×	×	✓
Lu et al. (J. Lu et al., 2019)	✓	✓	✓	✓	✓	✓	✓	✓	×	✓
Wares et al. (Wares et al., 2019)	✓	✓	×	×	✓	×	✓	×	×	✓
Khamassi et al. (Khamassi et al., 2018)	✓	×	✓	×	✓	✓	✓	×	✓	✓
Iwashita et al. (Iwashita & Papa, 2019)	✓	✓	✓	✓	×	×	×	×	✓	×
Ditzler et al. (Ditzler et al., 2015)	✓	×	✓	✓	×	×	✓	✓	✓	✓
Hoens et al. (Hoens et al., 2012)	✓	✓	✓	×	×	×	×	×	×	×
Our review	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

TABLE 2 Research questions related to concept drift handling in streaming data.

S. No.	Research questions
RQ1	What are the aspects of the concept drift problem?
RQ2	What are the types of concept drift detectors available, along with their features, pros and cons?
RQ3	What are the drift adaptation techniques available in the literature, along with their features and limitations?
RQ4	What are the available datasets for the evaluation of concept drift detectors?
RQ5	What performance metric(s) are used to evaluate the drift detectors?
RQ6	How the model drift monitoring process takes place?
RQ7	What is the current status of research in concept drift handling?
RQ8	What are the research trends in the field of concept drift?
RQ9	What are the research challenges in concept drift detection and adaptation in the streaming environment with labeled as well as unlabeled data?
RQ10	What are the possible future research options for handling concept drifts?

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1.2 | Our contribution

Following are the main contributions of this survey:

- Collection of the articles based on search strings and filtering them considering different criteria (Section 2).
- A thorough review of concept drift and its categorization based on speed, magnitude, and re-occurrence (Section 3).
- Review of the datasets utilized in literature for conducting experiments (Section 4).
- Study of state-of-the-art computational models and frameworks for detection and adaptation of concept drifts (Section 5 and 6).
- Review of the available performance assessment metrics for detection and adaptation techniques (Section 7).
- Discussion of future research scope for handling concept drifts (Section 8).

1.3 | Organizational structure of the manuscript

The rest of the paper is structured as follows: Section 2 explains the relevant article selection process and most searched research questions. Section 3 comprises of exploration of the concept drift and its types. Section 4 outlines the available data sets for experimental work. Section 5 discusses the drift detection techniques in streaming data. Section 6 covers the drift adaptation frameworks. Section 7 discusses the performance evaluation techniques available in the literature. Section 8 includes future directions in the area of concept drift detection in streaming data. The paper concludes with Section 9. Figure 1 represents the layout of the organization of the aforementioned section.

2 | CRITERIA FOR SELECTION OF RELEVANT ARTICLES

This review paper assists the researchers in grasping the fundamentals of concept drift detection and adaptation approaches in streaming data. This in turn identifies the most significant research problems of this domain. This section highlights the criteria for collection of research papers. After filtering the relevant literature for this survey, it can be said that the majority of the research papers are based on semi-supervised and supervised concept drift detection techniques and a very few of them focus on unsupervised techniques. There are a very few survey papers that discuss drift adaptation algorithms and performance evaluation metrics for drift detection mechanisms. However, this survey covers all the aforementioned topics.

2.1 | Information source

Predominantly, conference and journal research papers have been explored to extract related articles. Some popular databases such as Science Direct, Scopus, IEEE Xplore, ACM Digital Library, Springer and Wiley have been used to retrieve literature. The initial count of retrieved papers was enormous, that is, 9115. To identify relevant research papers and publications, various keywords and search strings mentioned in Table 3 were employed. Then, the process of screening was carried out, and the articles have been excluded with justifications such as (a) different title/abstract, (b) based on video streaming data, and (c) short papers, posters or other kinds of less contribution articles. The titles and abstracts of the articles between 2012 and 2022 included in the review have been examined further, and final papers have been examined after meeting the inclusion criteria. The strategy for filtering papers is mentioned in Figure 2, and the summary of no. of articles retrieved from different search databases is mentioned in Table 4 and graphically represented in Figure 3.

2.2 | Search criteria

It is necessary to organize the research in terms of comparison, intervention and results, therefore, having a good search string contributes to the purpose. The relevant papers were retrieved by integrating keywords from research questions

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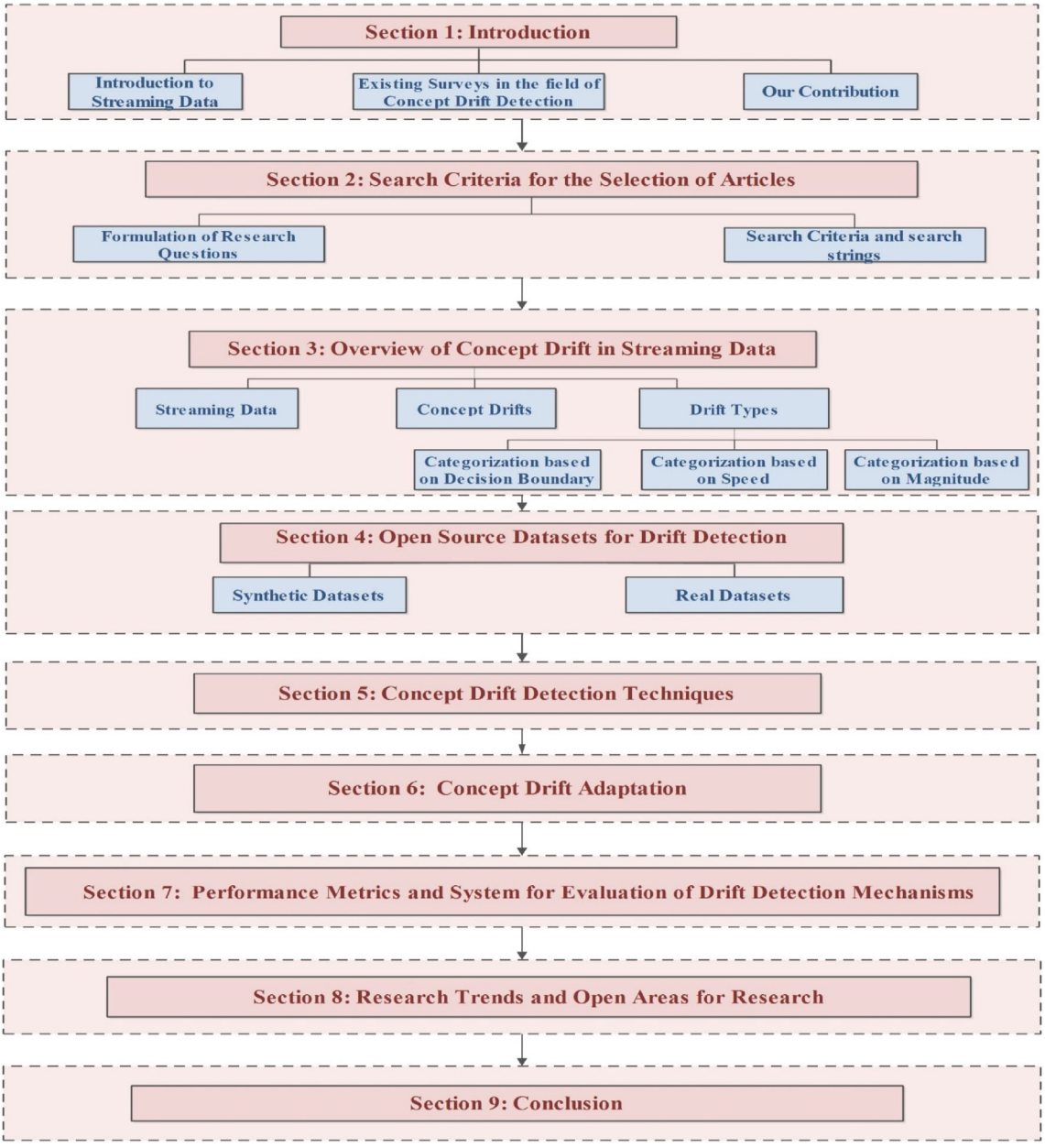


FIGURE 1 Layout of the manuscript.

TABLE 3 Search strings used for selection of relevant papers.

S. No.	Search strings
1.	"concept drift" AND ("detection" OR "Adaptation") AND ("streaming data" OR "data streams")
2.	("change detection") AND ("data streams" OR "streaming data")
3.	("data stream classification") AND ("concept drift")
4.	"Learning from concept drifts" AND ("data streams" OR "Streaming data")
5.	"drift adaptation" AND ("streaming data" OR "data streams")
6.	"detecting concept change in streaming data"
7.	"drifting streaming data"
8.	"Handling concept drift" AND "non-stationary environments"
9.	TITLE-ABS-KEY(("concept drift") AND ("Streaming data" OR "Data Streams"))

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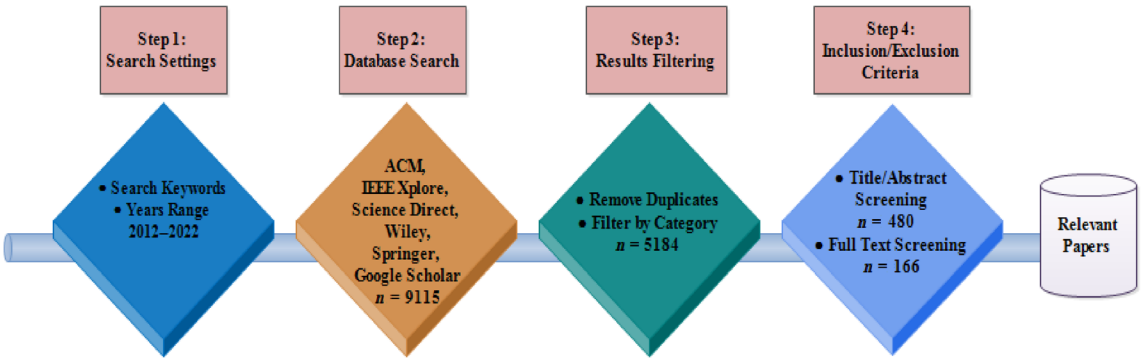


FIGURE 2 Search and selection criteria for relevant literature.

TABLE 4 The tally of articles obtained from various search databases.

***Search string*: Title, abstract: ("concept drift" AND ("detection" OR "adaptation")) AND ("streaming data" OR "data streams")**				
Search DB	**Total articles**	**Articles after screening**	**Eligibility**	**Inclusion**
Science Direct	41	36	23	9
IEEE Xplore	7144	4215	358	67
ACM	31	30	30	30
Springer	1814	845	27	25
Wiley	88	61	45	38
Total	9115	5184	480	166

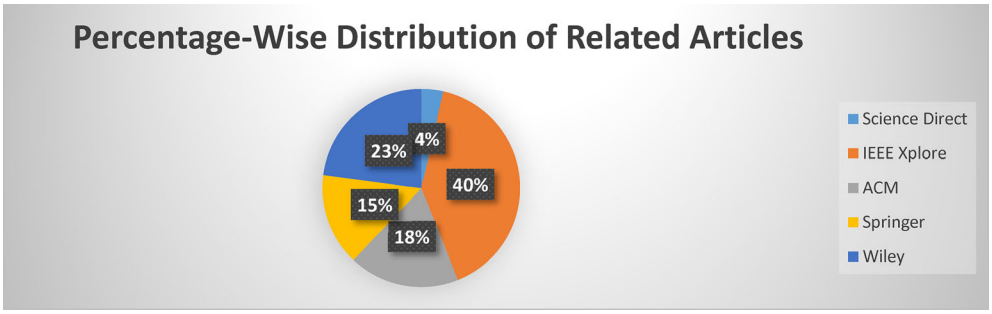


FIGURE 3 Distribution of concept drift related articles in different databases.

into a search string. Some of the search strings used to search the related literature from different pool of digital libraries are mentioned in Table 3.

3 | OVERVIEW OF CONCEPT DRIFT IN STREAMING DATA

3.1 | Streaming data

Data streams have specialized characteristics that set them apart from other data types. The data is commonly referred to as "batch data" in typical machine learning scenarios. The difficulties in processing data streams is that the arriving elements are exacerbated by their distinctive attributes.

The software platform designed for ingesting and processing massive amounts of streaming data from heterogeneous sources is known as streaming data architecture. The architecture consists of a stream input component that

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consumes data as soon as it is generated, persists it to store, and may include additional components depending on its application, such as tools for real-time computation, data manipulation, and analytics.

Data streams create a huge amount of data (terabytes to petabytes), which is most appropriately categorized as semi-structured and requires extensive pre-processing and Extract-Transform-Load (ETL) operations to make it usable. Researchers and businesses have started using tools such as Apache Spark, Storm, Kafka, S4 (Janardan & Mehta, 2017), and so forth for processing high-frequency data streams from distinct sources to get insights of the data in real-time. These computing systems assist business analysts to make valuable decisions. Moreover, these frameworks are open-source, and many tutorials are available to get started with them. There are various online machine learning algorithms available that run on streaming platforms such as Linear regression, Ridge regression, Logistic regression (Lin et al., 2014), LASSO, and so forth.

After the pre-processing task, the industries rely on the prediction or classification results. Streaming data solutions ought to be capable of managing voluminous data in real-time from a variety of sources while maintaining availability, scalability, and fault tolerance. In a data stream analytics system, the application takes the shape of continuous queries, and data is continuously ingested, evaluated, and correlated, leading to a stream of results. Streaming analytic applications often comprise of a series of operators linked together by streams. They must detect new data, develop models progressively, and determine whether new data deviates from model predictions. There are various recent surveys (Gehring et al., 2019; Malek et al., 2017) that have explained the stream learning platforms.

There are some applications of data streams in different sectors such as:

1. Finance:
   a. *Algorithmic trading*: Using real-time market data to execute high-frequency trades based on predefined algorithms.
   b. *Fraud detection*: Continuously monitoring transactions and user behavior to identify and prevent fraudulent activities in banking and financial systems.

2. Healthcare:
   a. *Remote patient monitoring*: Collecting and analyzing streaming health data from wearable devices and sensors to monitor patients' health conditions remotely.
   b. *Emergency response systems*: Utilizing real-time data from medical devices and emergency calls to facilitate quick responses to medical emergencies and disasters.

3. IoT:
   a. *Smart cities*: Leveraging streaming data from sensors embedded in urban infrastructure to optimize resource allocation and improve public services.
   b. *Industrial IoT (IIoT)*: Using streaming data from manufacturing equipment and machinery to support predictive maintenance and process optimization in factories.

4. Retail:
   a. *Inventory management*: Utilizing streaming sales data and supply chain information to optimize inventory levels and prevent stockouts.
   b. *Personalized marketing*: Analyzing streaming customer data to deliver targeted promotions and personalized recommendations in real time, enhancing customer engagement.

To tackle data distribution change, a number of researchers have come up with solutions in the past few years. A few have been mentioned in the following sub-sections.

3.2 | Concept drift

Concept drift is defined as the alteration in distribution of data with time due to various reasons such as model degradation or change in data trends. It takes place over time, and with variable frequency. A variety of symptoms precede concept drift, including the disappearance of past instances or the fact that their dispersion no more accurately reflects the class to which they belong.

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Concept drift can mathematically be stated by making an assumption that the learner model A is given a set of labeled instances (M_0, M_1, ..., M_t) at time t, where M_i is a n-dimensional feature vector and every data point has a known class label Z_i. The model assigns a class to an unlabeled instance M_{t+1}. It is assumed that the actual label Z_{t+1} and a unique testing data point M_{t+2} become accessible after the label is identified. At time t, the hidden function f yields an instance. The concept drift occurs when the aforementioned function f changes. Consider classification of instance M_{t+1}: it is required that the following information needs to be known to classify it optimally. The conditional probability of observing M_{t+1} given each class is $p(M_{t+1}|Z_i)$, and the prior probability of observing each class is $p(Z_i)$. The Bayes theorem then computes the probability that M_{t+1} is an instance of class Z_i using Equation (1).

$$p(Z_i|M_{t+1}) = \frac{p(Z_i) * p(M_{t+1}|Z_i)}{p(M_{t+1})}, \tag{1}$$

where $p(M_{t+1})$ is the probability for M_{t+1}. The drift can therefore happen because of either of the below mentioned key requirements in Bayes' theorem, as pointed out by Kelly et al. (1999):

a. $p(Z_i)$ is prone to vary (i.e., class priors).
b. $p(M_{t+1}|Z)$ (i.e., the proportion of the class) may vary.
c. The subsequent probabilities of class membership ($p(Z|M_{t+1})$) may alter.

Drawing an analogy to real-life scenarios,

- Manufacturing:
  Advertising budget, promotions, and client footfall might be seen in a model trained to estimate weekly sales. Taking an instance of an ice-cream parlor. However, model's accuracy might be affected due to seasonality. If ice-cream sales are more in the spring and summer due to seasonality, then examples from the months of winters may not correctly classify class labels. It is likely that although the models 'results indicate poor sales, the winter months may really see weaker sales because fewer individuals will be eating because of seasonality. Hence, once the concept drift has occurred, models need to be able to forget previous examples.
- Healthcare:
  In medical diagnosis, patient data may exhibit concept drift as disease patterns change, new treatments are introduced, or patient demographics shift. Detecting and handling concept drift ensures that diagnostic models remain accurate and reliable over time, improving patient outcomes.
- Financial markets:
  In stock market prediction, the behavior of stocks can change over time due to factors like economic conditions, news events, or regulatory changes. Detecting and adapting to concept drift is crucial for maintaining the accuracy of trading algorithms and minimizing financial risk.
- Natural language processing (NLP):
  In sentiment analysis or topic modeling applications, language usage and sentiment can change over time due to cultural shifts, news events, or emerging trends. Detecting and handling concept drift helps maintain the accuracy of NLP models, ensuring relevant and up-to-date insights.
- Environmental monitoring:
  Environmental monitoring systems collect data from sensors deployed in various ecosystems, which may exhibit concept drift due to climate change, natural disasters, or human activities. Detection and adaptation algorithms can improve environmental risk assessment, pollution control, biodiversity monitoring, and disaster response.

Key issues due to concept drift in streaming data:

- The classifier performance degrades with time due to concept drifts and gradually, it becomes outdated.
- It is challenging to monitor the evolving concept.
- It is difficult for the classifier to adjust to recent instances under non-stationary data distribution.

Significance of detection of concept drifts:

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- Improves the accuracy and reliability of predictive models.
- Enhances decision-making processes and business outcomes.
- Adapts to changing environments and evolving data distributions.
- Maintains a competitive edge in dynamic markets and industries.

3.3 | Types of drifts

The distribution of data streams is dynamic as they are continuous in nature. The data trends may change with time, and their distribution may differ. Such changes in the data distribution may be broadly categorized into two types on the basis of change in decision boundary, that is, (a) real concept drift and (b) virtual concept Drift. The categorization of drifts can also be done on the basis of the speed of concept change, that is, by dividing them into five major types, namely (i) abrupt (Sudden), (ii) gradual, (iii) incremental, (iv) recurring and (v) blip. A blip can also be termed as an outlier since it denotes a fast or sudden change (or rare event) in a concept. However, it can be ignored from the categorization of drifts as it falls under abrupt category. Further, the drifts can also be categorized into two types on the basis of magnitude, that is, local concept drift and global concept drift. The overview of categorization of concept drift is depicted in Figure 4.

3.3.1 | Categorization of drifts based on decision boundary

a. Real concept drift:

In the case of real concept drift, the decision boundary $P(Y|X)$ changes, that is, there is a variation in the conditional probabilities. Such drifts alter the problem's actual decision boundaries lowering classifier performance directly as depicted in Figure 5b.

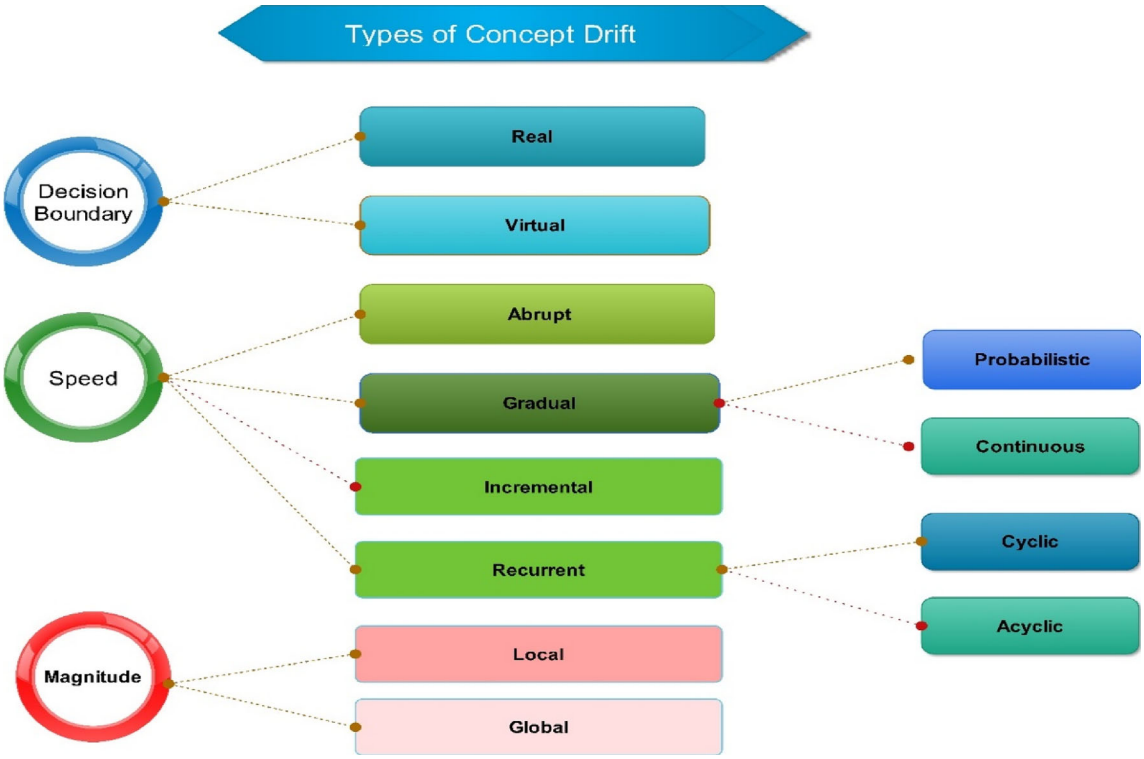


FIGURE 4 Categorization of concept drifts on different criteria.

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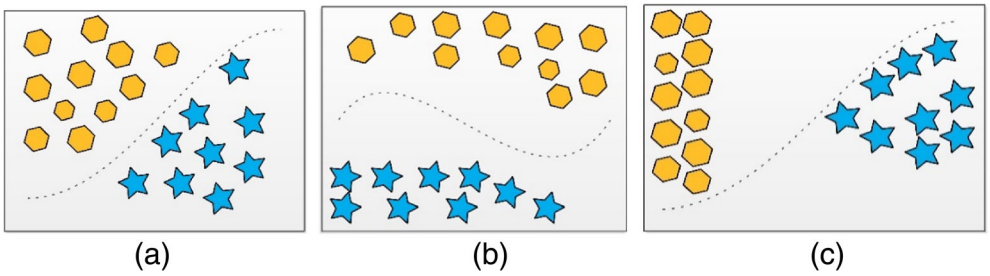


FIGURE 5 Different forms of concept drift in streaming data. (a) Original data; (b) Real drift; (c) Virtual drift.

b. Virtual concept drift:

In the case of Virtual Concept Drift, an unconditional probability distribution P(x) varies can be seen in Figure 5c. These kind of drifts have received far less interest among researchers since they do not modify the problem's actual decision bounds. They can, however, influence classifier performance by changing the adequacy of the decision boundaries learned by the classifiers.

There is currently no comprehensive knowledge of the variations in the effects of these variants of drift on classifier. Consequently, current streaming algorithms consider virtual drifts in the similar way as real drifts (Oliveira et al., 2021).

3.3.2 | Categorization of drifts based on speed

The number of epochs essential for a new concept to replace an old one is known as drift duration, also known as drifting time or drift breadth (Ghomeshi et al., 2019) and speed is inversely proportional to drifting time (L. L. Minku et al., 2010). It signifies that a faster speed corresponds to smaller time steps. Concept drifts can be classified into abrupt, gradual and recurrent, depending on their speed.

i. *Abrupt (sudden) drift*: It happens when a novel concept takes the place of an outdated one. As the novel concept rapidly replaces the outdated one, this drift variant immediately degrades the model performance.
ii. *Gradual drift*: This type of the drift takes place when the drift duration is prolonged. Since it generates an interval of complexity in steady states, this type of drift is more difficult to identify. Gradual Drift is classified into two categories, that is, Probabilistic and continuous gradual drift.
iii. *Recurrent drift*: A new concept or a recurring concept that has occurred at previous time steps of the data stream may develop again due to concept drift. For instance, On Black Friday Sale, people who shop moderately throughout the year display interesting patterns where the weekend mobility varies from that on working days. This is another classification of drifts that aims to elaborate previously occurred drifts that may occur again in the future or have trends to occur after a specific season. Recurrent drifts can have two types of behaviors, that is, cyclic and acyclic. The appearance of recurring concepts might be sudden or gradual, and they can affect the instance space locally or globally. During the transition phase, several different characteristics of drift might be detected.
iv. *Incremental drift*: It is also known as continuous drift, and entails a long drift duration. In this case, the progression from concept P to concept P′ is gradual. The time steps taken by an evolving concept to replace an outdated concept entirely is the speed, or duration, of concept drift.

3.3.3 | Categorization of drifts based on magnitude

It is the degree of variation caused by the drift. Based on the severity standard, concept drifts can be classified as local and global drifts (Khamassi et al., 2018).

i. *Local concept drift*: It refers to events in certain areas of the instance space. As a result, considering the entire instance space, it can be seen that only a portion of it is impacted by the drift. Local drift can be considered noise

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caused by uncertainty due to which the model becomes volatile. As a result, the model must efficiently differentiate between noise and local changes.

ii. *Global concept drift*: It affects the whole instance area therefore, the Global Concept Drift can be effortlessly spotted. In this situation, the distinction between the outdated and current concepts is more apparent, and the drift is possible to be identified immediately.

4 | FREQUENTLY USED DATASETS RELATED TO CONCEPT DRIFT

In this section, we review some real-world and synthetic datasets that have been used in literature to evaluate and model concept drift scenarios. As the start and end points of the drifts are known in advance, the synthetic datasets allow us to examine how the approaches cope with the various types of drift present in the datasets.

4.1 | Synthetic datasets

Please refer URL: https://www.dropbox.com/sh/cx58kb5yctdimln/AADPP__VynluSkHssXwC4DNxa?dl=0

1. SINE1: The challenge of determining arrangement of coordinates, which are illustrated by two variables, in reference to the curve $y = \sin x$ is posed by Sine Data. Positive points are those below the curve in the first scenario. The classification is inverted post every drift. The values of every coordinate are evenly dispersed over the [0, 1] range. To maintain a steady learning environment comparable to those outlined in Baena-García et al. (2006), positive and negative instances are altered.
2. SINE2: This dataset has two characteristics (p and q) that are evenly dispersed in the range [0, 1], similar to SINE. The occurrences that fall below curve are categorized as positive, and the remaining cases are categorized as negative, according to the function $q = 0.5 + 0.3 \sin(3p)$. A drift occurs every 20,000 times, and the classification is then reversed (Mahdi et al., 2020).
3. CIRCLES: This dataset exhibits a gradual drift as well as circumstances where there is no noise. It contains four categorization functions, each of which is controlled by four circles (four concepts). Instances are categorized based on their location: if they are inside the circle specified by the circular function, they are positive; otherwise, they are negative. The incremental modification is accomplished by altering the circle's center and radius sizes (Gama et al., 2004).
4. STAGGER: This dataset contains negligible noise and abrupt drifts. The instances present in the dataset have three conceptual properties, particularly size, color and shape. If size = small and color = red, examples are positive in the primary concept. The instances in the secondary concept are defined by their color, which is green, or their shape, which is circular. In the tertiary concept, instances are classified as medium or large in size (Schlimmer & Granger, 1986).
5. SEA: This concept generator stimulates real, abrupt concept drifts with three real-valued attributes in the range [0, 10]. The stream contains 60,000 instances, 15,000 for each concept with distinct thresholds of concept function. A threshold function to identify 0 to a class instance is *appropriate_characteristic1* + *appropriate_characteristic2* > *Threshold* (Iwashita & Papa, 2019).
6. Rotating Hyperplane: The position and location of the hyperplane can be modified swiftly by adjusting the weights. Hence, hyperplanes are utilized to model time-varying concepts (S. Huang & Dong, 2006).
7. Checkerboard: The canonical XOR problem, which resembles a spinning checkerboard, yields a non-Gaussian data set. This dataset comprises of 20,000 instances and 2 features. The rotation shows the complexity of this fairly simple-looking problem. This is because the angle and position of the decision boundaries vary dramatically within a few epochs (Elwell & Polikar, 2011).
8. Random RBF: This dataset is generated using RandomRBFDriftGenerator present in the MOA framework. It consists of 10,000 instances with 4 categories and 4 features (Bifet, Holmes, Pfahringer, Kranen, et al., 2010).
9. GAUSS: This dataset contains abrupt drifts along with noisy instances and domain R × R having two related features. Center [0,0] and standard deviation = 1 are set to locate positive instances with normal distribution. For negative instances, center [2,0] and the value of standard deviation is set to 4. The classification is reversed as soon as drift is observed (Gama et al., 2004).
10. MIXED: This dataset is noiseless and contains abrupt drifts. The instances have four attributes, that is, two of Boolean type x, y and the other two are numeric between 0 and 1. The instance is categorized as positive if it satisfies

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the condition, that is, $x, y, z < 0.5 + 0.3 \times \sin(3\pi x)$. The classification is reversed as soon as drift is observed (Gama et al., 2004).

11. Agrawal: The hypothetical loan application is described by this generator, which was first proposed by Agrawal et al. (1993). It has six integer attributes and three categorical attributes. There is a perturbation factor for numeric characteristics that causes the true value to be shifted by adding an offset.

4.2 | Real-world datasets

1. ELEC: The electricity dataset is a real-world data set derived from the Electricity Market in NSW, Australia. Harries (1999) was the first to describe this data set. Prices in this electrical market are not predetermined and might fluctuate depending on market dynamics. The dataset collected is between the range of May 1996 and December 1998 and contains 45,312 samples. The class label indicates how the price has changed in relation to a 24-h moving average. Only the class level is presented for the price fluctuations on a 1-day average, eliminating the influence of relatively long market trends.
2. SPAM: Email message datasets are used to determine whether or not they are unsolicited messages. ECUE spam datasets, for example, are two datasets, each containing over 10,000 emails gathered over a span of around 2 years and are related to a single account.
3. Poker hand: This dataset is quite popular and is based on the challenge of identification of the hand in a Poker game. It comprises 1,025,010 instances depicting all potential poker hands, with every instance being a hand made up of five cards drawn from a standard 52-card deck. Two qualities (suit and rank) are assigned to each card in hand. Each hand is described by 10 characteristics (*UCI machine learning repository: Poker hand data set*, n.d.).
4. CoverType: There are 54 features in this dataset that describe different forms of forest cover. It has 581,012 entries, that elaborate seven different forest cover categories, according to the US Forest Service's Resource Information System (RIS) (Pesaranghader et al., 2018).
5. Weather: The National Oceanic and Atmospheric Administration of the United States has accumulated weather data from over 9000 meteorological stations (Ueno et al., 1999). A range of features (temperature, pressure, wind speed, etc.), as well as indicators for precipitation and other weather-related occurrences, are included in daily measurements. The labels of the class are based on the binary indicator(s) provided for each daily reading of rain, with a total of 18,159 daily readings.
6. Shuttle: This dataset has numerical type of 58,000 instances and nine attributes. Nearly, the amount of data that belongs to class 1 is 80%.
7. Sensor: It is a real data stream containing information such as humidity, temperature, light and voltage, which is collected using 54 sensors installed in Intel Berkeley Research Lab (Janardan & Mehta, 2017).
8. NSL-KDD database: This data set was proposed to address some of the flaws of KDD'99 data sets, as described in Tavallaee et al. (2009). This is a valuable benchmark dataset to be used for research while examining various intrusion detection systems as it does not contain redundant values. There are 41 attributes for each instance and a total of 25,192 instances distributed into 23 classes.
9. Adult: This dataset is based on the loan information to predict whether an individual earns over 50 K annually. There are 14 attributes of both categorical and numerical type and 48,842 instances in this dataset (Kohavi & Becker, 1996).
10. KDDcup99: This dataset consists of TCP connection records gathered from LAN network traffic at MIT Lincoln Labs. It has 23 labels, 494,021 instances and 41 attributes. Stolfo et al. (2000) created this data collection based on data from the DARPA'98 IDS evaluation program (Lippmann et al., 2000).
11. SEER Breast Cancer: This breast cancer patient dataset was collected from the NCI's Surveillance, Epidemiology, and End Results (SEER) Program's November 2017 update, which shares insights on population-based cancer statistics. The study comprised female patients diagnosed from 2006 to 2010 with breast cancer (Teng, 2019) (Table 5).

5 | CONCEPT DRIFT DETECTION TECHNIQUES

The change in underlying data is due to a shift in the statistical characteristics of the target variable. Drift detection can be accomplished in three steps. In the first step, the instances are employed to improve the learner that forecasts the

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TABLE 5 Summarized description of real-world datasets.

Dataset	#Instances	#Attrs	#Classes	Link
ELEC	1 M	10	10	https://www.openml.org/d/151
SPAM	10 K	57	2	https://archive.ics.uci.edu/ml/datasets/spambase
Pokerhand	1,025,010	10	10	https://archive.ics.uci.edu/ml/datasets/Poker+Hand
Covertype	581 K	53	7	https://archive.ics.uci.edu/ml/datasets/covertype
Weather	18, 159	8	2	https://users.rowan.edu/∼polikar/nse.html
Shuttle	58 K	9	7	https://archive.ics.uci.edu/ml/datasets/Statlog+(Shuttle)
Sensor	2 M	5	54	http://www.cse. fau.edu/xqzhu/stream.html
NSL-KDD	25,192	41	23	https://www.unb.ca/cic/datasets/nsl.html
Adult	48,842	14	2	https://archive.ics.uci.edu/ml/datasets/adult
KDDCup99	494,021	41	23	https://datahub.io/machine-learning/kddcup99
SEER Breast Cancer	4024	15	–	https://ieee-dataport.org/open-access/seer-breast-cancer-data

goal values. The data samples are then examined for concept drift in the next phase. Current data instances are predicted further if there is no drift in the data samples. If concept drift is discovered, the analysis of drift takes place, followed by drift adaptation mechanisms in the final step. It refreshes the learning model using current data instances during the adaption process. Concept drift detection is achieved in this manner in the streaming environment outlined in Figure 6.

There are a variety of ways to keep track of concept drift, as listed below:

a. The probability distribution of data is checked to detect concept drift because it may alter with time due to the factors such as distribution, noise, outliers, sensor delay, and so forth.
b. It is possible to check the occurrence of concept drift. It is essentially done by keeping track of the similarities and differences between various sample features or qualities.
c. The classifier's accuracy suffers as a result of concept drift. Consequently, it might be one of the measurements used to detect drift in a stream.
d. To determine the presence of concept drift, one of the input attributes that can be used is the timestamp. The timestamp function can be applied to a single sample or a group of samples (C. Te Li et al., 2016).

There are mainly three types of techniques that have been developed to address concept drift:

1. Adaptive base learners.
2. Training set modifying learners.
3. Ensemble.

5.1 | Adaptive base learners

This is the simplest method to tackle the problem of concept drift detection from a conceptual standpoint. Such learners can react dynamically to incoming training data that conflicts a previously learnt concept. Drift adaptation can take different forms depending on the base learner used. This section examines several adaptive base learners that researchers have developed in the recent years.

The decision tree is one of the oldest and most extensively used in the area of concept drift. The most frequent variation of the decision tree used is C4.5 (Kustiyahningsih et al., 2021). Very fast decision tree (VFDT) is a version of the decision tree learning technique that dealt with creation of decision trees from streaming input (Brzezinski, 2010). To develop decision trees in streaming data, Hoeffding limits (Moore, 1993) are employed in VFDT. In the context of data streams, the authors proved that setting Hoeffding bounds for a data subset can help in selecting the same split attribute with high confidence as using complete data. As a consequence, trees can be generated from streams which are closely related to those constructed with whole data. Hoeffding bounds (also known as additive Chernoff bounds

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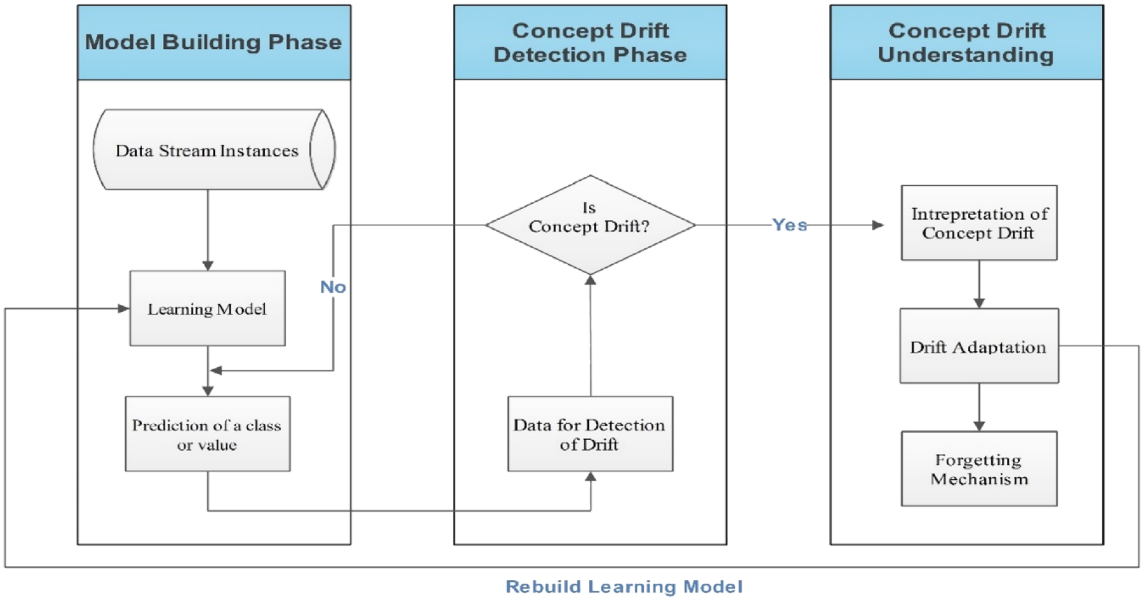


FIGURE 6 Main framework for concept drift detection and adaptation.

(Chernoff, 1952)) is said to be a quantitative approach for calculating confidence bounds on a distribution's mean. The Hoeffding bound asserts that, given n instances of random variable x with range r and observed mean of μ_o, the real mean of the variable is $\mu_o - \varepsilon$ with probability $1 - \beta$, where ε is given in Equation (2):

$$\epsilon = \sqrt{\frac{r^2 \ln(1/\beta)}{2n}}. \tag{2}$$

For streams that experience concept drift, many improvements to VFDT were done. Hulten et al. (2001) used VFDT to build the concept-adapting very fast decision tree (CVFDT). In CVFDT, a sliding window of instances is stored in short-term memory. Each node's relevant statistics are updated, and the Hoeffding bounds are recalculated after a certain number of fresh instances arrive. When a finer splitting attribute is discovered, the sub-tree concludes that the concept drift is detected and builds a subtree. The method further expects more instances to justify that the newly generated sub-tree shows better quality as compared to the initial, after which the original is replaced.

In contrast to fixed window in CVFDT, Hoeglinger and Pears (2007) presented a concept-based window as an alternative to CVFDT. Instead of refreshing the window considering time, they update it concerning concepts. Trees are developed only from the leaves to achieve this. Instances are preserved until the window gets full, at which point inefficient leaves are "fused" with their parent nodes and all the instances linked with the leaf are deleted. To minimize the overall loss of information for the tree, leaves are chosen to be recombined.

Bifet and Gavaldà (2009) presented new approaches recently: the Hoeffding window tree (HWT) and the Hoeffding adaptive tree (HAT). With two key differences, HWT is however, identical to CVFDT. The first difference is that instead of waiting for a predetermined number of instances, HWT builds alternative (sub)trees right away, allowing the algorithm to adapt to concept drift immediately. Furthermore, instead of waiting for a certain number of instances to refresh a sub-tree, HWT prefers to update when there is evidence to support the new sub-tree's enhancements. HWT has a notable benefit over CVFDT where concept drift can be identified more efficiently.

Another advantage of HAT over CVFDT is that it offers guaranteed performance. Buntine (2020) suggested option trees as an extension to normal decision trees, that were subsequently investigated by Kohavi and Kunz (1997) and Arora et al. (2023). Traditional decision trees only have one way to get from the root to the leaf node, which is where predictions are generated. However, with option trees, a different node variant is introduced called an option node, that branches the route along various split nodes. HOTs are extended in adaptive Hoeffding option trees (AHOTs) (Bifet, Holmes, Pfahringer, Kirkby, & Gavaldà, 2009). Each leaf in AHOTs is given an exponential, weighted average estimator with a fixed decay of 0.2.

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The CD3 approach suggested by Black and Hickey (2003) is an alternative to CVFDT, in which the authors recommended the development of decision tree with a time-stamp component added for every instance. Analyzing each path to the leaves and identifying if the time-stamp was an essential attribute in the newly built tree, the algorithm may detect concept drift after learning the training data. The rule is said to be acceptable, and the data is maintained if the time-stamp is significant and denotes the most recent period. If the time-stamp originates from an earlier period, the rule is declared invalid, and the previous occurrences are deleted. Finally, the rule is even said to be valid in the absence of time-stamp component.

5.2 | Training set modifying learners

Altering the training set observed by the classification technique is another typical way to tackle concept drift. Windowing (leveraging only a subset of previous instances) and assigning weights to the instances are the most prevalent strategies. Modification techniques are generally classifier agnostic, which the advantage of modification technique over adaptive base learners. Therefore, the majority work is done to modify the training set that focuses on the identification of weight instances used to build a classifier rather than on how to build the complete classification system. The following sections discuss training set modification techniques and distinguish between complete learning algorithms and only detection procedures.

5.2.1 | Window techniques

The primitive technique for updating the training set via window based technique is to simply preserve a finite number of the most recent occurrences (i.e., a "window" over the modern instances in the data stream) (Dongre & Malik, 2014). This naive approach has various flaws, the most significant one is the inability to establish the optimum window size for each problem a priori. Many different methods have been proposed in an attempt to address this limitation. Adaptive Windowing (ADWIN) given by Bifet and Gavaldà (2007) which recognizes the various types of changes using sliding windows. Concept drift can be detected if the difference in means between two sub-windows is higher than the threshold. The processing time is directly proportional to the window size, that is, it increases with an increase in the size of the window. As a result, the authors have offered ADWIN2, a developed version that meets the memory and temporal constraints. While maintaining roughly the same performance assumptions as ADWIN, ADWIN2 preserves a window of length W with O(logW) memory and time updations. The minimal time and memory feature of ADWIN2 makes it to be used in another way: a learning algorithm that can construct several instances of ADWIN2 to keep the statistics (counts, averages, entropies, etc.) from which it constructs the up to date model.

Another approach that uses a single classifier is the concept adapting very fast decision tree (Hulten et al., 2001). It added the feature of concept drift detection in VFDT algorithm. A sliding window is also used to keep the classifier up to date with latest instances.

The authors (Du et al., 2014) presented ADDM (Adaptive Window-based Drift Detection Method), a window-based algorithm that dynamically determines the window size. It identifies concept drift by keeping track of the window's entropy. When the entropy is equal to one, it signals a concept drift. Seven datasets, having various types of concept drift, were used to test ADDM, that resulted in high accuracy.

Liu et al. (2017) proposed a fuzzy windowing method called FW-DA concept drift adjustment. When the test statistics of the current and previous windows differ significantly, the suggested algorithm reports a drift.

Nishida and Yamauchi (2007b) presented STEPD algorithm, which takes into account the accuracy of two recent and old windows. Drift is identified if the difference between the two windows is considerable as determined by a statistical test.

Another approach known as Fast Hoeffding Drift Detection Method (FHDDM), proposed by Pesaranghader and Viktor (2016) monitors the probabilities of correct predictions on the sliding window. It analyses the highest and current probability and examines change if the deviations are equal to or greater than the threshold.

An unsupervised strategy that detects concept drift by measuring the modifications in the feature space leveraging a discriminative classifier with a sliding window knows as D3 was designed by Gözüaçık et al. (2019). It is a straightforward strategy that can be applied to any existing classifier that does not have a built-in drift adaptation mechanism.

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An intelligent approach mainly developed for real-time data derived from wireless sensor networks is known as Adaptive sliding window random decision Tree (ASWRT) (Almalki & Abdullah, 2019). The findings of this study were primarily based on the concept of data segmentation. Segmentation is an essential part of extracting meaningful patterns from time series data, which can help by spotting trends and changes in prediction. ASWRT involves majority of segmentation tasks on the basis of mean and variance. For classification purposes, a random decision tree was implemented for an incremental tree.

Lazarescu et al. (2004) proposed employing two windows (called small and medium) instead of a single window, with the small window having a set size of S and the medium window having a minimum size of 2S. As fresh instances arise, the algorithm continually updates the small window. Whereas, the medium window is altered when concept drift is detected. To capture the new concept, the medium window is set to size 2S if concept drift is recognized over a certain amount of observations. The medium window can then increase to a maximum size of M once the concept has settled.

Sayed-Mouchaweh et al. (2011) in their work presented a technique for tracking concept drift for the improvement of classifier performance. This method employs an adaptive time window size for drift detection based on its dynamics (slow, moderate, or fast). The main objective is to retrain the classifier with a significant number of patterns associated with environmental changes.

Bach and Maloof (2008) proposed paired learners that pair a static online learner with a dynamic one to deal with concept drift. A static learner makes predictions based on their entire experience, whereas a reactive learner makes predictions based on their current experience.

5.2.2 | Weighting techniques

Weighting techniques are frequently implemented in classification, where the relative worth of each instance is used. When compared to windowing strategies, instance weighting allows the learning algorithm to have more accurate control over how instances are included in the model than merely being declared as "present" or "not present."

An exponentially weighted moving average (EWMA) chart proposed by Ross et al. (2012) is used to monitor the misclassification rate of a streaming classifier in order to detect concept drift. This method is modular and may be used in conjunction with any underlying classifier to give an extra layer of concept drift detection.

Pesaranghader et al. (2018) introduced the McDiarmid Drift Detection Method (MDDM), which detects concept drift using McDiarmid's inequality (Wallstrom, 2017). The MDDM method works by sliding a window over model performance and assigning weights to window elements. To underline the relevance of the most recent entries, higher weights are provided to them. The detection technique compares a weighted mean of elements inside the sliding window with the greatest weighted mean observed so far as examples are processed. A considerable discrepancy between the two weighted averages, which is upper-bounded by the McDiarmid inequality, indicates that the concept drift has occurred. All the mentioned techniques are summarized in Table 6.

5.3 | Ensemble techniques

The use of ensemble learning has proven to be a successful approach since it improves predictive power, facilitates drift management, and is relatively simple to use in practical applications (Gomes et al., 2017; Krawczyk & Cano, 2018). Existing studies (Lv et al., 2019) have attempted to give weights or thresholds to each classifier utilizing a variety of methodologies, but they have underperformed. There are mainly three categories of ensemble methods, that is, chunk-based, online and hybrid ensemble. The chunk-based approach for a data stream with class imbalance was initially put forth in Peng et al. (2021). The suggested method balances the classes by under-sampling the majority class samples while adding the minority class samples from all previous chunks. Then, classification technique is implemented using an ensemble framework that performs similar to bagging. However, it struggles with memory issues while maintaining historical data. There are several other chunk-based and online ensemble techniques implemented in literature which are discussed in Tables 7 and 8, respectively.

Lv et al. (2019) proposed a robust classifier for learning data streams by utilizing the online bagging ensemble technique. They implemented a reliable coding technique based on error-correcting output codes in their classifier, as well as an influential online resampling mechanism for the training cases. A dynamic updating approach based on categorization performance is used to lessen the redundant updating operations and boost learning effectiveness.

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TABLE 6 Summary of existing drift detection algorithms.

Algorithm	Acronym	Type	Technique(s)
Concept adaptive very fast decision tree (Kumar et al., 2015)	CVFDT	Window-based	Creates another sub-tree to locate optimal attribute at root-node
Paired learners (Bach & Maloof, 2008)	PL	Window-based	Utilizes two learners: Stable and reactive
Adaptive windowing (Bifet, 2009)	ADWIN	Window-based	Keeps distinct windows. Each window represents one of the *n* dimensions of the data
SEED (D. T. J. Huang et al., 2014)	SEED	Window-based	It determines stream volatility and cut point for drift detection
Drift detection method (Gama et al., 2004)	DDM	Statistical method	It operates according to binomial distribution theory
Early drift detection method (Baena-García et al., 2006)	EDDM	Statistical method	Uses error rate of distance and performs well for slow, gradual drift
Statistical test of equal proportions (Nishida & Yamauchi, 2007b)	STEPD	Statistical method	The accuracy of recent data chunk is computed and compared with overall accuracy from the start of the stream. Further, a chi-squares test is employed to check for deviation
Fast Hoeffding tree drift detection method (Pesaranghader & Viktor, 2016)	FHDDM	Window and Hoeffding's inequality	Calculates difference between the maximum probability of correct predictions and the most recent probability of correct predictions
Exponentially weighted moving average (Ross et al., 2012)	EWMA	Statistical method	Monitors the misclassification rate of streaming classifier using EWMA chart
Fisher test drift detector (Cabral & Barros, 2018)	FTDD	Statistical method	Applied in data mining to discover the inter-dependence of attributes
Plover algorithm (Mello et al., 2019)	PA	Statistical method	Checks the deviation of the mean of new data to past data and compares it with the threshold value. It finds out the weak learners in classification or regression. The data stream is divided using a windowing mechanism and evaluates the parameters such as mean, variance, skewness, and kurtosis
Streaming ensemble algorithm (Street & Kim, 2001)	SEA	Block-based ensemble	The classifiers are clubbed together into a static size ensemble incorporating a heuristic replacement technique
Accuracy weighted ensemble (Dalhoumi et al., 2015)	AWE	Block-based ensemble	To eliminate the poorly performing classifiers, AWE uses mean square error (MSE) to choose '*n*' best classifiers for the construction of completely new ensemble
Accuracy updated ensemble (Brzezinski & Stefanowski, 2014)	AUE	Block-based ensemble	The online classifiers are implemented to enable the independent learning models to be improved directly rather than tuning the weights
Dynamic weighted majority (Kolter & Maloof, 2007)	DWM	Incremental ensemble	Each classifier in the ensemble is given some weight. The overall prediction is made by combining weights and prediction of each classifier
Cumulative sum (Page, 1954)	CUSUM	Sequential analysis	If the incoming data's mean differs from zero by a substantial amount, it outputs an alarm. The test's input can be a residual from any predictor, such as the Kalman filter's prediction error
Page-Hinkley (Page, 1954)	PH	Sequential analysis	It is a variant of CUSUM and is sequential adaptation of the detection issue. It operates by calculating the observed values and their means up to the present time

Abbasi et al. (2021) proposed ElStream that uses both real and synthetic data to detect concept drift using ensemble and traditional machine learning approaches. EIStream uses the majority voting method, allowing just the best classifier to cast a vote.

Elwell and Polikar (2011) designed an algorithm known as Learn++ which is a nonstationary data distribution technique and a classification method that is based on ensemble techniques. It learns from the data batches next to

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TABLE 7 Chunk-based ensembles for stationary data streams.

Technique	Description	Limitation(s)
Learn++ (Polikar et al., 2001)	Builds distinct neural network models upon every input data chunk, then uses majority voting to aggregate the results	Inefficient for processing large datasets
Ada.Boost RAN-LTM (Kidera et al., 2006)	A robust neural network classifier for incremental learning integrates AdaBoost.M1 and Resource Allocating Network with Long-Term Memory	Not Applicable for potentially unbounded streams
Growing NCL (F. L. Minku et al., 2009)	• Formed by combining a group of accurate and heterogeneous neural networks for co-training • Suitable for a continued trade-off between the forgetting rate and responsiveness to the inconsistent incoming input	Low generalization and does not take advantage of employing multiple ensembles for learning new data
Bagging++ (Zhao et al., 2010)	• Used to create new models from incoming data chunks, an enhancement over Learn++	It does not produce precise values for the classification or regression model; instead, it reinforces its final prediction based on the mean predictions of subgroup trees
Dynamic updated ensemble (DUE) (Z. Li et al., 2020)	• The ensemble interacts with various types of concept drift because of periodic updates from DUE • Final choice of evaluating events is based on the weighting vote of a defined set of best components	When a classifier's weight drops down to a set threshold, DUE discards it, limiting the accurate response to recurring drifts
Adaptive chunk-based dynamic weighted majority (Y. Lu et al., 2020)	ACDWM uses an ensemble framework and dynamically weights the various classifiers based on how well they classify the current chunk of data	Appropriate mainly for tackling issues with binary imbalance
Dynamic weighted majority for imbalance learning (DWMIL) (Y. Lu et al., 2017)	• Maintains stability in streams that are not drifted and rapidly adapt to the changing concept • It is entirely incremental, indicating that previous data is not required to be stored • It requires a single threshold value and maintains a small number of classifiers to ensure high performance	Ensuring that the current chunk of data has enough minority instances is challenging
Chunk-adaptive restoration (CAR) (Kozal et al., 2022)	This framework is suitable for classification of block-based data stream. It adjusts the chunk size of data to detect concept drift	The performance gain is not visible for other types of drifts except abrupt drift, but it still achieves acceptable results
Streaming ensemble algorithm (SEA) (Street & Kim, 2001)	• SEA preserves the classifiers that properly label the instances that are almost unresolved by the ensemble • Both overfitting and diversity maintenance are possible • The ensemble's predictions are made based on majority voting	It is likely for old classifiers to surpass new classifiers, which could impede the integration of new concepts
Accuracy weighted ensemble (AWE) (H. Wang et al., 2003)	AWE assigns weight to every classifier in the ensemble according to how well they predict the most recent training batch. A particular form of the mean square error is utilized (that enables handling of component classifier prediction probabilities)	During abrupt drifts, AWE's pruning technique occasionally deletes an excessive number of component classifiers, henceforth, reducing the classification accuracy by an excessive amount
Accuracy updated ensemble (AUE) (Brzeziński & Stefanowski, 2011)	• Weighting classifiers with non-linear error functions are a novelty that helps to encourage more accurate classifiers • The most recent classifier is always given the highest value of weight as it reflects the most recent data distribution	Ensemble diversity problem exist in case where many components with similar examples are updated
Weighted aging ensemble (WAE) (Woźniak, 2013)	The weight of a specific classifier is determined by its accuracy and the amount of time it spends inside an ensemble	The accurate comparison evaluation of the recognition effectiveness against state-of-the-art approaches would be hampered by obvious data peeking that results from fine-tuning
Cost-sensitive continuous ensemble kernel learning method (CCEKL) (Chen et al., 2024)	Proposed an ensemble approach to address the algorithm sensitivity to the initial kernel width	Partial and delayed labeling challenges are unaddressed in both balanced and imbalanced data stream environments

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TABLE 8 Online ensembles for stationary data streams.

Technique	Description	Limitation
OzaBag (Oza & Russell, 2001)	• Gradual progression of the basic bagging algorithm • Uses a Poisson ($\lambda = 1$) distribution to show how the bootstrapping procedure might be modified for use in an online environment	Its high computational cost might deter adoption in some circumstances
ASHT (Bifet, Holmes, Pfahringer, & Gavaldà, 2009)	• If the ASHT tree's number of split nodes exceeds the maximum value, it deletes some nodes to make the tree smaller • It has a value for the maximum number of split nodes or size after one node splits	Less ensemble diversity
LevBag (Bifet, Holmes, & Pfahringer, 2010)	Improve randomization by applying output detection codes and increasing resampling to maximize the performance of bagging	Causes a loss of a model's interpretability
OzaBoost (Oza, 2005)	A static size collection of classifiers trained on the instances so far received is maintained by this ensemble. Updates are made to each classifier sequentially using every new instance	
Ultra Fast Forest of Binary Trees (UFFT) (Gama & Medas, 2005)	• Performs online learning by using a collection of Hoeffding trees • Solely use the split criterion to perform binary classification tasks	Based on the split criterion, it does not support drift detection
Hoeffding Option Tree (HOT) (Gama, 2010)	• Enables every training instance to modify a set of option nodes instead of a single leaf • Performs similar to a group of weighted classifiers and standard Hoeffding Trees	It gives a predictive performance better than VFDT at the cost of a significant memory increase
Ensemble of Online Extreme Learning Machines (EOS-ELM) (Lan et al., 2009)	The initial heterogeneity of the pool is achieved via a stochastic training approach, and it is a straightforward combination of online randomized neural networks.	No mention of how the ensemble's diversity is verified to be maintained
Dynamic Weighted Majority (DWM) (Kolter & Maloof, 2007)	• Allows the ensembles to highlight the classifiers most likely to be accurate at a certain time • The incoming training samples are used to train all classifiers incrementally • Feasible to add a new classifier or eliminate current classifiers in order to react to concept drift faster	May not perform well in stationary conditions
Adaptive Classifiers Ensemble (ACE) (Nishida & Yamauchi, 2007a)	• Batch classifiers trained on previously used instances saved in a buffer are used in addition to an online classifier to learn new training examples • If the most accurate batch classifier's accuracy is outside the confidence interval, ACE determines that there has been concept drift	ACE needs incremental instance storage in order to generate batch classifiers; however, this problem is simply fixed by substituting an online learning algorithm for the buffer
Diversity for dealing with drifts (DDD) (L. L. Minku & Yao, 2012)	• Based on the notion of diverse ensembles that help detect concept drifts • When drift is detected, predictions using the high diversity ensemble become active for learning	Employing more than one ensemble for applications with extreme time limitations can make this strategy heavier
Online bagging with ADWIN drift detector (ADWINBagging) (Bifet,	• The ADWIN algorithm has been integrated with the online bagging method as a change	

(Continues)

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TABLE 8 (Continued)

Technique	Description	Limitation
Holmes, Pfahringer, Kirkby, & Gavaldà, 2009)	detector and an estimator for the weights of the boosting approach • As soon as a difference is noticed, the underperforming classifier is replaced by a new one	Some concerns of runtime and memory consumption have been overcome by its base classifier variants

it. It may detect cyclic or recurring drift as well as a constantly changing rate of drift. It employs weighted majority voting and modifies the accuracy in real-time based on the present and past environment.

P. Wang et al. (2022) proposed a weighted concept drift detector based on incremental ensemble learning. The detector takes into account not only the outcome of the prediction but also any changes in the predictor's stability as a result of concept drift.

Alippi et al. (2009) extended their kNN-based method to study under slow, incremental drift in adaptive weighted kNN. In their earlier work (Alippi & Roveri, 2008a, 2008b), they proposed preserving all occurrences in the "new" knowledge base and removing them from the "old" one.

6 | CONCEPT DRIFT ADAPTATION

It is often necessary for a prediction model to monitor the concept drift due to the dynamic nature of streaming data. Due to the concept drift condition, the data distribution changes over time in the stream, and the model needs to be able to adapt to this distribution change. The most effective techniques for adaptation are based on ensemble methods. It helps the classifier become more accurate and segments big problems down into smaller ones (Krawczyk et al., 2017).

To deal with an adaptive mechanism in non-stationary distributed contexts, detection and adaptation is necessary to keep track of change. The adaptation mechanism is essential in a variety of situations, including work status, age variations, seasonal patterns, changes in consumer behavior, changes in market dynamics, and so forth. There are two types of adaptation techniques:

- *Active strategy*: If the drift is explicitly identified, then the data distribution is altered.
- *Passive strategy*: The model changes continuously over time (Ditzler et al., 2015).

The active learning strategy for concept drift relies on a change detection system that, when appropriate, sets off an adaptation process to update or create a new classifier in response to the change. The analysis of the extracted features monitors the stationarity of the estimated $f_t(c)$, whereas the study of the classification error aims to detect variations in the estimated $f_t(z \mid c)$. The change detector seeks to assert that "there is a change in the process P" by examining features extracted from the data generation process for change detection purposes and analysis of the classification error (evaluated over labeled samples). Only when a change is detected, the adaptation phase starts that regenerates the classification model. The "detect & react" strategy is another name for adaptive algorithms (Zang et al., 2014). When a change is identified, outdated knowledge is discarded by the classifier and adjusted to the new environment.

In prediction scenarios with gradual drifts and recurring concepts, passive techniques have proven highly effective (Elwell & Polikar, 2011). Dealing with abrupt drift can be accomplished using active procedures (Alippi et al., 2009). Additionally, while active approaches have been demonstrated to be equally effective in online environments, passive ones are typically better suited for batch learning (Alippi et al., 2009; Bifet & Gavaldà, 2007; Gama et al., 2004).

Two general categories of streaming systems adapt to changes in data distribution: blind and informed (Gama et al., 2014). Blind methods consider the most recent findings without specifically identifying concept drift. A typical illustration is training a model using a sliding window of recent observations. Older observations are ignored because it is thought that the distribution from which new observations would be drawn will be best described by the most recent observations that were captured in the window. Another illustration is the Concept-Adapting Very Fast Decision Tree (CVFDT), which creates a decision tree progressively and uses the Hoeffding constraint to choose when to add a new branch. New branches are built using recent data to replace outdated branches. There is a trade-off between stability and adaptability that determines how soon observations are forgotten. Longer observation retention increases the stability of the classifier, which is crucial in stationary settings but can be slow to respond to change.

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Informed methods explicitly identify distributional changes. By promptly deleting outdated observations, explicit detection enables a quicker response to concept drift while maintaining more crucial data when no concept drift is found (Klinkenberg & Renz, 1998). A classifier's error rate can be significantly elevated to indicate a shift in the distribution of the data. The Drift Detection Method (DDM) (Gama et al., 2004), Early Drift Detection Method (EDDM) (Baena-García et al., 2006), and Adaptive Windowing (ADWIN) (Bifet, 2009) are some drift detection techniques that are based on this concept. A commonly used informed strategy for building a classifier involves progressively updating it while the data distribution remains stable, without any detected drift. Recurring concept approaches assume that concepts could repeat across the data stream and that previously established correlations might become more applicable. The ability to store a list of previously used classifiers is a feature common in various techniques (Gama & Kosina, 2014; Gonçalves & Barros, 2013). A saved classifier may be utilized to achieve higher efficiency than rebuilding a new classifier if a concept recurs.

Further, the concept drift adaptation techniques can be categorized into two main abstract ideas, that is, memory and forgetting data. Adaptive learning systems are made up of modular components that can be merged and permuted.

7 | PERFORMANCE METRICS AND SYSTEM FOR EVALUATION OF DRIFT DETECTION MECHANISMS

The evaluation systems are a crucial component of algorithmic learning. In Gama et al. (2014), some evaluation techniques for learning with concept drift are described. The existing research is enriched by analyzing the frameworks in terms of three criteria, that is, validation, parameters, and statistical significance. Each evaluation is followed by an explanation of its computation equation and intended use. A learning algorithm selects which data examples to use as the training and testing set, this process is known as validation methodology. Three approaches unique to the evaluation of learning algorithms that recognize concept drift, that is, holdout, prequential, and controlled permutation are outlined in Table 9. Holdout adheres to the criterion when evaluating a learning algorithm at time t: the holdout set precisely represents the same concept at a moment t. It is, however, only used with artificial datasets that have concept drift times that are predetermined. A common evaluation method for data streams is prequential. Each data instance is utilized twice: firstly, to test the learning algorithm and secondly, to train it. The benefit of this technique is that it makes the best use of the data that is already available and does not require knowledge of the timings of concept drift. Based on the accumulated sum of the loss function between the predicted and observed labels, the prequential error is calculated as given in Equation (3):

$$E = \sum_{i=1}^{n} t(y_i, y). \tag{3}$$

A landmark window (interleaved-test-then-train), a sliding window, and a forgetting mechanism are three prequential error rate estimates. Data instances that were close to one another in time initially must remain close after a permutation, according to controlled permutation (Gama et al., 2013; Zliobaite, 2011).

Traditional accuracy measurements, such as precision or recall in data retrieval, mean absolute scaled error for regression, or root mean square error for some task of recommendation systems, could be chosen as evaluation metrics

TABLE 9 Some evaluation methods and their corresponding metrics.

Method of evaluation	Metric(s)
Prequential Evaluation	accuracy, kappa, kappa_t, kappa_m, true_vs_predicted, precision, recall, f1, g_mean
Holdout Evaluation	Accuracy
CD-MOA	RAM-hours, Cohen's Kappa Statistic
Mean Time Ratio (MTR)	Encapsulates the ratio between MTFA and MTD metrics, Mean time to detection
No Change Detector	Outputs false positive change after every 60 instances

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for datasets featuring concept drift. Additionally, it is important to look into another metrics such as Ram-Hour(s), Kappa Statistic, Kappa-Temporal, Prequential AUC, precision-recall curve, False detection rate (FDR), and so forth.

8 | RESEARCH TRENDS

The increase in the amount of real-time data, opens the scope to create new techniques leveraging huge systems. Change detection and adaptation in the streaming environment are the key areas of research. Further, the study of concept drift phenomena is evolved in variety of fields related to data stream mining. Some of the fields include:

1. Concept shift: The change in underlying data generator, results in concept change. As a result, whenever the context of streaming data changes, it is said to be a conceptual shift. For instance, Insulin resistance is an example, in which a person's medical category shifts from insulin non-resistance to resistance.
2. Evolution of the concept: Concept evolution describes the items that the classifier has not previously learned. It explains the novel idea of data instances. For instance, in the network-based intrusion detection system, the class represents various types of threats, and new distribution data instances are created throughout the computational process.
3. Mapping the drift: The mapping in data stream mining specifies the type of drift for a certain timespan. As opposed to concept drift, where the algorithms for detecting drift attempt to determine the drift's exact location. There are some scenarios addressed through drift mapping, that is, How the various drift methods function in different types of concept drift, Identifying the drift, and the speed at which the detector finds the significant change (Agrahari & Singh, 2021).
4. Handling recurrent concept drifts: Concept drift is predicted to reoccur over time and across diverse items. For a certain subgroup of entities, seasonal effects with ambiguous periodicity would be typical (e.g., in the forecast of food supply) (Zliobaite et al., 2012). To better manage recurrent concept drift, additional relevant data may be made available or latent contexts might be extracted from the predictive features (Gama & Kosina, 2011). Additionally, it appears that transfer learning between the various contexts could be a promising research area (Goyal et al., 2023).

8.1 | Open areas for future work

1. Constructing models that process data streams by doing single pass over the data, eliminating the requirement of data storage.
2. Implementing drift detection methods that give better precision to the real drift point.
3. The study of the concept drift in data stream mining still faces a serious challenge of imbalanced classification.
4. An interesting unexplored area in streaming data is Zero-Shot-Learning which means learning to identify the concept by just having a description of it. This description might be a set of language tags or an arithmetic field. This would be extremely relevant to the streams with significant concept drift or streams with concept evolution.
5. Some earlier works proposed the potential concept of “self-labelling.” However, it is impractical to use it in the streaming environment. A classifier's high confidence level might not be justified in periods of change (particularly with genuine drift or concept evolution), and the iterative training process might take too much time to be useful in a data stream.
6. Events must first be detected in order to be analyzed, hence combining “Event History Analysis” with techniques for event detection is a significant problem.
7. Research on drift detection should not only concentrate on precisely defining the time of drift occurrence, but need to include some information about drift intensity and areas. These details might be used to improve adaptability of concept drift.
8. The idea that using accuracy as a performance indicator for concept drift detectors is inadequate. The development of a new statistical evaluation metric is one potential area of future research.
9. Making concept drift detection techniques capable of tackling anomalies such as temporal dependence is also in its infancy stage and can be a good direction for future research.
10. When handling drifts, the heterogeneity of the ensemble members might be a hot study area.
11. Mitigating concept drift under missing value is still an unaddressed problem that requires extensive research.

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9 | CONCLUSION

The analysis of unbounded, potentially fast-moving data streams is a significant field of study. There is a hypothesis that concepts within dynamic data streams are subject to alter which contributes toward the change in statistical properties of the data. The problem of concept drift in streaming data that is discussed in this survey require solutions that can be implemented in a limited amount of time and memory. In this survey, concept drift and its types, state-of-the-art detection and adaptation algorithms, synthetic and real-world datasets and performance evaluation criteria of the available algorithms is discussed comprehensively. The novelty of this survey article is that it categorizes types of drifts based on decision boundary, speed and magnitude. Further, the survey also provides the categorization of concept drift detection algorithms into Adaptive base learners, training set modifying learners, and ensemble techniques. Based on the research questions specified in the earlier section of this paper, a total of 166 articles were studied for drift types, detection, adaptation and performance evaluation techniques. This article also discusses current and anticipated trends in the concept drift domain of streaming data and further suggests some open areas for future work.

There are some survey limitations that may be addressed in future work such as:

a. Limited practical implementation insights: While the survey discusses various detection and adaptation algorithms for concept drifts, it may lack detailed insights into the practical implementation challenges and considerations faced by the researchers when applying these techniques in real-world scenarios.
b. Evaluation metrics: Although the survey highlights performance metrics used to evaluate concept drift detection methods, it may not provide an exhaustive analysis of the strengths and limitations of each metric. The choice of evaluation metrics and their appropriateness for different applications could be further explored.

AUTHOR CONTRIBUTIONS

Shruti Arora: Data curation (lead); formal analysis (lead); methodology (lead); visualization (lead); writing – original draft (lead); writing – review and editing (lead). **Rinkle Rani:** Supervision (equal); validation (equal). **Nitin Saxena:** Supervision (equal); validation (equal).

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

OPEN RESEARCH BADGES

This article has earned an Open Data badge for making publicly available the digitally-shareable data necessary to reproduce the reported results. The data is available at https://github.com/alipsgh/data-streams.

DATA AVAILABILITY STATEMENT

The datasets generated during and/or analyzed in the current study are available from the corresponding author on reasonable request.

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